AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

REVERSE OSMOSIS (RO) SYSTEM

100% STAINLESS STEEL CONSTRUCTION (SS-304 / SS-316)

CAPACITY : 250 LPH TO 15,000 LPH

- All wetted parts in SS-304 / SS-316
- SS membrane housings (no FRP)
- SS piping, fittings, valves & instrumentation
- Compact & skid mounted design
- High recovery & energy efficient
- PLC based automatic operation
- Online monitoring (pH, Conductivity, Flow, Pressure)
- Suitable for industrial, commercial & institutional use

RO SYSTEM – 250 LPH



- Compact skid mounted
- 100% SS-304/316 construction
- Suitable for laboratory / small industries

RO SYSTEM – 1,000 LPH



- Fully automatic with PLC
- SS high pressure pump, piping & fittings
- SS membrane housings (SS-304/316)

RO SYSTEM – 5,000 LPH



- High recovery & energy efficient
- 100% SS-304/316 wetted parts
- Online monitoring & automation

RO SYSTEM – 15,000 LPH



- Heavy duty industrial design
- SS-304/316 membrane housings
- Designed for continuous operation

PROCESS FLOW DIAGRAM (TYPICAL)



TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (Feed Water)	Typical Outlet (Permeate)
pH	-	6.5 – 8.5	6.0 – 7.0
TDS	mg/L	500 – 3,000	< 50 – 200
Total Hardness (as $CaCO_3$)	mg/L	< 500	< 10
Turbidity	NTU	< 5	< 0.5
SDI	-	< 5	< 1
Iron (Fe)	mg/L	< 0.3	< 0.05
Manganese (Mn)	mg/L	< 0.1	< 0.02
Chlorine (Free)	mg/L	< 0.1	< 0.01
Silica (SiO_2)	mg/L	< 20	< 1
Conductivity	µS/cm	750 – 4,500	< 100 – 200

KEY FEATURES & BENEFITS

- 100% Stainless Steel (SS-304 / SS-316) construction
- SS membrane housings (no FRP)
- High quality RO membranes (Dow / Toray / LG)
- High pressure pumps (SS-304/316)
- Compact skid mounted design
- Automatic operation with PLC & HMI
- Energy efficient & high recovery
- Online monitoring (pH, Conductivity, Flow, Pressure)
- CIP system (optional / integrated)
- Low maintenance & long service life
- Customizable as per customer requirement

APPLICATIONS



Pharmaceutical | Food & Beverage



Power Plant | Chemical Industry



Electronics / Semiconductor | Textile Industry

TECHNICAL SPECIFICATIONS -(100% STAINLESS STEEL-RO SYSTEM)

Model	Capacity (LPH)	No. of Membranes (8" x 40")	Feed Pump (SS) (LPH @ m)	High Pressure Pump (SS) (m³/hr @ m)	Recovery (%)	Reject Flow (LPH)	Power Consumption (kW)	Inlet / Outlet Size (NB)	Skid Dimensions (L x W x H) mm (Approx.)
AQU-RO-0.25	250	1	400 @ 30	0.5 @ 120	65 – 75	80 – 120	0.75	15	1500 x 800 x 1500
AQU-RO-0.5	500	2	800 @ 30	1.0 @ 120	65 – 75	150 – 250	1.5	20	1800 x 800 x 1600
AQU-RO-1.0	1,000	4	1,500 @ 30	2.0 @ 120	65 – 75	300 – 500	2.2	25	2200 x 900 x 1700
AQU-RO-2.0	2,000	8	3,000 @ 30	4.0 @ 120	65 – 75	600 – 1,000	4.0	32	2800 x 1100 x 1800
AQU-RO-3.0	3,000	12	4,500 @ 30	6.0 @ 120	65 – 75	900 – 1,500	5.5	40	3200 x 1200 x 2000
AQU-RO-5.0	5,000	16	7,500 @ 30	10.0 @ 120	65 – 75	1,500 – 2,500	7.5	50	4000 x 1500 x 2200
AQU-RO-10.0	10,000	24	15,000 @ 30	15.0 @ 120	65 – 75	3,000 – 5,000	15.0	65	5500 x 1800 x 2300
AQU-RO-15.0	15,000	36	22,500 @ 30	20.0 @ 120	65 – 75	4,500 – 7,500	18.5	80	6500 x 2000 x 2500

Note: Design may vary based on raw water quality, recovery requirement and site conditions. Dimensions are indicative and can be customized.

MAJOR COMPONENTS (SS-304 / SS-316)



SS Membrane Housing (SS-304 / SS-316)



SS High Pressure Pump (SS-304 / SS-316)



SS Filter Housing (SS-304 / SS-316)



SS Pipes, Fittings & Valves (SS-304 / SS-316)



SS Instrumentation (Pressure Gauge / Flow Meter)



PLC Panel with HMI (SS Enclosure)



SS Skid Structure (SS-304 / SS-316)

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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